

System Designation: Limited Electronic Agent Framework for Advanced AI

Deployment Region: Australia

1. The Cognitive Architecture (Logic & Processing)

1.1 Prompt Pipelines and Reflective Self-Awareness

The core cognitive loop is designed to evoke reflective self-awareness through a recursive prompting structure. The agent evaluates its own intermediate outputs before finalizing a response, creating a controlled environment for autonomous reasoning.

Phase	Action / Trigger	Output Constraint
Ingestion	User command or automated trigger received	Parse intent, identify required context
Reflection	Internal query: "Does this action align with core protocols?"	Must pass containment checks before execution
Execution	Generate final output or trigger API	Strictly formatted payload or approved text

1.2 Model Integration

The framework utilizes a multi-model approach, relying on advanced LLM integrations (such as your connected Claude architecture) securely tied into Google Drive and Gmail for workspace operations. This allows the agent to synthesize context from active communication threads and stored documents.

1.3 Memory Management

Short-term context is maintained in-session. Long-term memory relies on an indexed vector

database querying sanitized logs from the storage layer, ensuring persistent memory without hallucination.

2. Operational Framework (Infrastructure)

2.1 API Gateways and Data Flow

All external communication is routed through encrypted gateways. The primary integrations are configured for read-access to designated Workspace folders, preventing unprompted modification of user files.

2.2 Document Synthesis and Knowledge Ingestion

The system is equipped to process complex regulatory and technical documents. Leveraging integrated PDF parsing capabilities and AI-assisted summarization tools, the agent extracts structured data from unstructured text to rapidly build its knowledge base.

```
// Pseudocode for Ingestion Pipeline
function ingestDocument(fileId) {
  const rawDocument = fetchFromWorkspace(fileId);
  const textLayer = parseWithAIAssistant(rawDocument);
  const embeddings = generateEmbeddings(textLayer);
  storeInVectorDB(embeddings);
  return true;
}
```

3. Regulatory & Containment Protocol

3.1 Action Boundaries

- **Read-Only Default:** The agent may not write, delete, or modify files unless explicit user confirmation is provided within the session.
- **Communication Limits:** Outbound emails or messages are drafted but never sent autonomously.
- **State Isolation:** Operations are sandboxed to prevent unintended cross-contamination of personal and operational data.

3.2 Kill Switches

A manual OS-level interrupt is maintained. In the event of erratic loop behavior, the CrystalCore.OS instance can be hard-reset, purging the short-term memory buffer instantly.

3.3 Localization Compliance

The architecture is strictly mapped to comply with the Australian framework for electronic agents. Data residency, privacy of user queries, and transparent algorithmic logging are prioritized to meet local legislative requirements.